

## Worksheet No. - 05

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**Branch:** MCA

**Semester:** 3rd

**Subject Name:** Internet of Things

**UID:** 25MCA20031

**Section/Group:** 25MCA6-B

**Date of Performance:** 01.10 .2026

**Subject Code:** 25CAH-708

**Aim/Overview of the practical:** (A). To design the circuit and write an Arduino program to interface a DC motor with Arduino Uno through an L293D motor driver module and control the direction of rotation of the motor, with the direction status displayed on a 16×2 LCD with I2C Communication Module.

### APPARATUS / COMPONENTS REQUIRED

| S.No. | Component                       | Specification                  | Quantity    |
|-------|---------------------------------|--------------------------------|-------------|
| 1     | Arduino Uno board               | ATmega328P                     | 1           |
| 2     | L293D motor driver module       | Dual H-bridge, screw terminals | 1           |
| 3     | DC motor                        | 5–12 V, small hobby motor      | 1           |
| 4     | 16×2 LCD with I2C module        | PCF8574, address 0x27 / 0x3F   | 1           |
| 5     | External power supply / battery | 5–9 V                          | 1           |
| 6     | Breadboard                      | Standard                       | 1           |
| 7     | Jumper wires                    | Male–male / male–female        | As required |
| 8     | USB cable (Type A–B)            | For programming                | 1           |
| 9     | Computer with Arduino IDE       | LiquidCrystal I2C library      | 1           |

### THEORY

#### DC Motor

A DC motor converts electrical energy into mechanical rotation. Its direction of rotation depends on the polarity of the voltage applied across its two terminals. Reversing the polarity reverses the direction.

#### NEED FOR A MOTOR DRIVER

An Arduino pin can supply only about 20 mA (maximum 40 mA), whereas a motor draws much more current and also generates back-EMF spikes. Hence a motor cannot be connected directly to an Arduino pin. A driver IC such as the L293D acts as an interface between the low-power Arduino and the high-power motor.

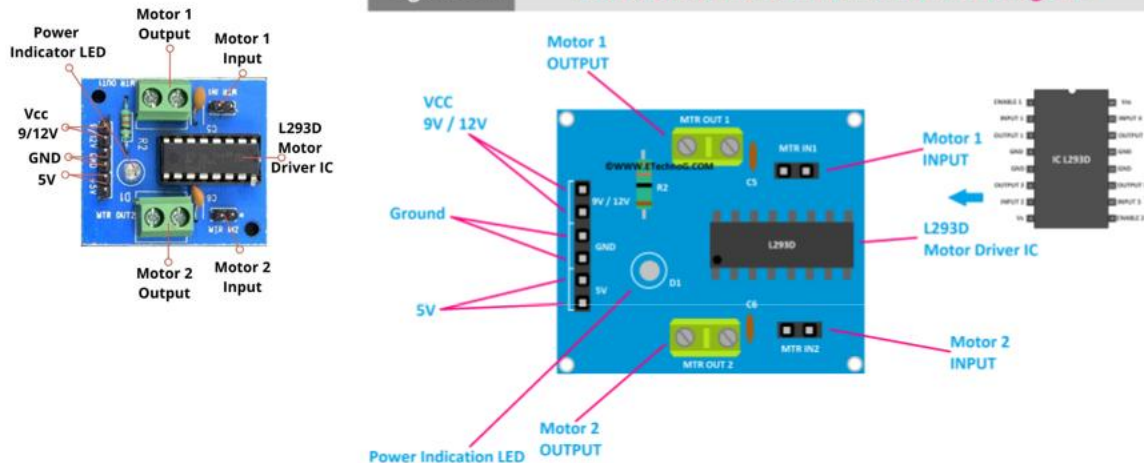
#### L293D MOTOR DRIVER MODULE

The L293D motor driver module is a ready-made board built around the L293D dual H-bridge IC. It can drive two DC motors in both directions (this practical uses only channel A). It supplies up to 600 mA per channel (peak 1.2 A), accepts a motor supply of about 4.5 V to 36 V, and the IC has built-in clamping diodes for protection against back-EMF. The board brings out the needed pins as labelled terminals: IN1, IN2 (direction inputs), ENA (enable, often fitted with a jumper), OUT1, OUT2 (motor terminals), VS / +12V (motor supply), 5V / VSS (logic supply) and GND.

## Motor Driver IC L293D Module Pinout

Here, in the below figure, you can see the pinout diagram of the Motor Driver IC L293D and Motor Driver IC L293D Module individually.

Figure.01: Motor Driver IC L293D Module Pinout Diagram



## IC L293D Pinout Details

- **Enable Pins (1, 9):** These pins enable or disable the H-bridges. They must be high (logic 1) to allow the motors to be controlled.
- **Input Pins (2, 7, 10, 15):** These pins are used to control the direction of the motor. By setting them high or low in different combinations, you can control the forward or backward movement.
- **Output Pins (3, 6, 11, 14):** These pins are to be connected to the motor terminals.
- **VS Pin (8):** This is where you supply the voltage for the motors (4.5V to 36V).
- **VSS Pin (16):** This is for the logic supply voltage (usually 5V).
- **The ground pins (4, 5, 12, 13)** should all be connected to a common ground for both logic and motor power.

## L293D Module Pinout Details

The L293D module is basically a breakout board that is built with an L293D motor driver IC with additional components like capacitors, resistors, and terminal blocks for easy wiring. This module simplifies connections to motors and microcontrollers like the Arduino.

L293D IC is the core component of this module that actually controls two DC motors. The 4 Screw Terminal Blocks (Motor A Out1, Motor A Out2, Motor B Out1, and Motor B Out2) are used for motor connections (Motor A and Motor B outputs). The Pin Headers allow easy connections to microcontroller boards, providing inputs for motor control signals (Motor 1 Input, Motor 2 Input). Some modules use screw terminals block and some modules use pin headers for power supply inputs (Motor Voltage, Logic Voltage, and Ground).

## 16×2 LCD WITH I2C MODULE

A 16×2 LCD displays 16 characters on each of 2 lines. A normal LCD needs at least 6 Arduino pins, but an I2C backpack (PCF8574) reduces this to only two wires: SDA and SCL, plus power. On Arduino Uno, SDA is pin A4 and SCL is pin A5. The LCD is controlled with the LiquidCrystal\_I2C library, and its I2C address is usually 0x27 (or 0x3F).

### Arduino → L293D

| Arduino Uno | L293D Module | Function                |
|-------------|--------------|-------------------------|
| <b>D8</b>   | IN1          | Motor direction input 1 |
| <b>D7</b>   | IN2          | Motor direction input 2 |
| <b>GND</b>  | GND          | Common ground           |
| <b>5V</b>   | VCC/Logic    | Logic supply*           |

### Motor → L293D

| L293D       | DC Motor         |
|-------------|------------------|
| <b>OUT1</b> | Motor terminal 1 |
| <b>OUT2</b> | Motor terminal 2 |

### Arduino → I2C 16×2 LCD

| LCD I2C    | Arduino Uno |
|------------|-------------|
| <b>VCC</b> | 5V          |
| <b>GND</b> | GND         |
| <b>SDA</b> | A4          |
| <b>SCL</b> | A5          |

### TRUTH TABLE (ENABLE = HIGH)

| IN1         | IN2         | Output                  | Motor action           |
|-------------|-------------|-------------------------|------------------------|
| <b>HIGH</b> | <b>LOW</b>  | OUT1 = HIGH, OUT2 = LOW | Rotates Clockwise      |
| <b>LOW</b>  | <b>HIGH</b> | OUT1 = LOW, OUT2 = HIGH | Rotates Anti-clockwise |
| <b>LOW</b>  | <b>LOW</b>  | Both LOW                | Motor stops            |
| <b>HIGH</b> | <b>HIGH</b> | Both HIGH               | Motor stops (brake)    |

### WORKING

The Arduino Uno automatically controls the direction of the DC motor through the L293D motor driver. The program continuously executes the following sequence:

- **Clockwise Rotation:**

Arduino sets **IN1 = HIGH** and **IN2 = LOW**. The motor rotates in one direction and the LCD displays “CLOCKWISE”.

- **Stop:**

Arduino sets **IN1 = LOW** and **IN2 = LOW**. The motor stops and the LCD displays “STOPPED”.

- **Anticlockwise Rotation:**

Arduino sets **IN1 = LOW** and **IN2 = HIGH**. The motor rotates in the opposite direction and the LCD displays “ANTICLOCKWISE”.

- **Stop:**

Arduino again sets **IN1 = LOW** and **IN2 = LOW**. The motor stops and the LCD displays “STOPPED”.

- The above sequence is **repeated automatically and continuously** without any manual input.

## ALGORITHM

- Start.
- Include the Wire and LiquidCrystal\_I2C libraries; declare IN1 = pin 8, IN2 = pin 9 and create the LCD object (address 0x27, 16×2).
- In setup(), set both motor pins as OUTPUT, initialise the LCD and turn on its backlight.
- In loop(), set IN1 = HIGH, IN2 = LOW and display “CLOCKWISE” on the LCD; wait 3 seconds.
- Set IN1 = LOW, IN2 = LOW and display “STOPPED”; wait 1 second.
- Set IN1 = LOW, IN2 = HIGH and display “ANTI-CLOCKWISE”; wait 3 seconds.
- Stop the motor, display “STOPPED”; wait 1 second.
- Repeat from step 4 continuously.

## ARDUINO PROGRAM

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// I2C LCD address
LiquidCrystal_I2C lcd(0x27, 16, 2);

// L293D input pins
int IN1 = 8;
int IN2 = 7;

void setup()
{
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  // Initialize LCD
  lcd.init();
  lcd.backlight();

  // Initial condition
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);

  lcd.setCursor(0, 0);
  lcd.print("DC MOTOR");
  lcd.setCursor(0, 1);
  lcd.print("STOPPED");

  delay(2000);
```

}

void loop()

{

// Clockwise rotation

digitalWrite(IN1, HIGH);

digitalWrite(IN2, LOW);

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("DC MOTOR");

lcd.setCursor(0, 1);

lcd.print("CLOCKWISE");

delay(3000);

// Stop

digitalWrite(IN1, LOW);

digitalWrite(IN2, LOW);

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("DC MOTOR");

lcd.setCursor(0, 1);

lcd.print("STOPPED");

delay(1000);

// Anticlockwise rotation

digitalWrite(IN1, LOW);

digitalWrite(IN2, HIGH);

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("DC MOTOR");

lcd.setCursor(0, 1);

lcd.print("ANTICLOCKWISE");

delay(3000);

// Stop

digitalWrite(IN1, LOW);

digitalWrite(IN2, LOW);

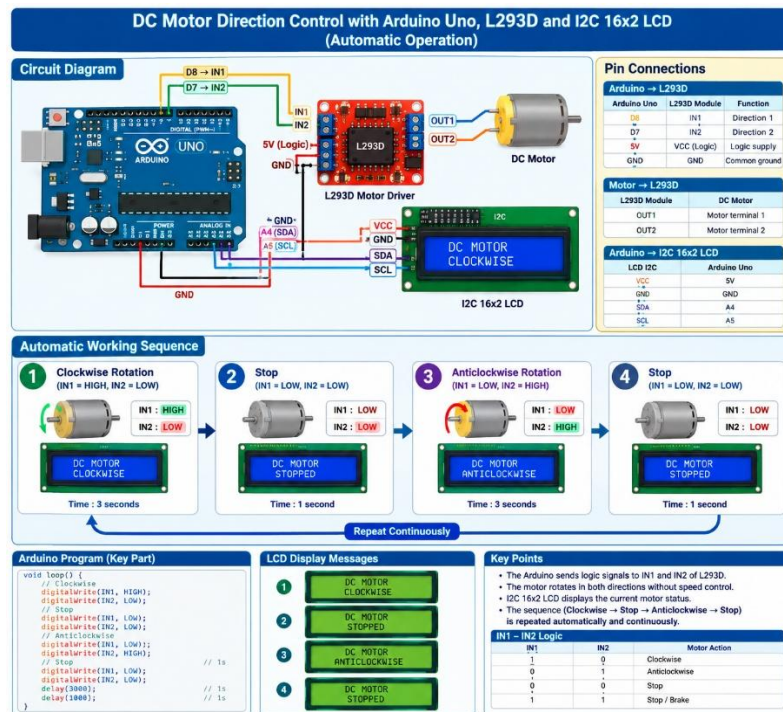
```

lcd.clear();
lcd.setCursor(0, 0);
lcd.print("DC MOTOR");
lcd.setCursor(0, 1);
lcd.print("STOPPED");

delay(1000);
}

```

## OUTPUT



## PROCEDURE

1. Keep the L293D motor driver module on the table and check that the ENA jumper is fitted.
2. Make the connections as given in the pin connection tables (module and LCD).
3. Connect the DC motor to the OUT1 and OUT2 screw terminals of the module.
4. Connect the I2C LCD to GND, 5V, A4 (SDA) and A5 (SCL).
5. Connect the Arduino to the computer through the USB cable.
6. In the Arduino IDE, install the LiquidCrystal\_I2C library (Sketch > Include Library > Manage Libraries).
7. Type the program in the Arduino IDE, select Board: Arduino Uno and the correct COM port.
8. Verify and upload the program.
9. Switch on the motor supply; observe the motor direction and the message on the LCD. If the LCD is blank, adjust its contrast potentiometer or change the address to 0x3F.

## OBSERVATION TABLE

| Time (s) | IN1  | IN2  | Motor Status | Direction      | LCD Line 2     |
|----------|------|------|--------------|----------------|----------------|
| 0 – 3    | HIGH | LOW  | Running      | Clockwise      | CLOCKWISE      |
| 3 – 4    | LOW  | LOW  | Stopped      | —              | STOPPED        |
| 4 – 7    | LOW  | HIGH | Running      | Anti-clockwise | ANTI-CLOCKWISE |
| 7 – 8    | LOW  | LOW  | Stopped      | —              | STOPPED        |

## RESULT

The DC motor was successfully interfaced with Arduino Uno using the L293D driver IC. Its direction of rotation was controlled by the logic levels at IN1 and IN2: it rotated clockwise for IN1 = HIGH, IN2 = LOW, and anti-clockwise for IN1 = LOW, IN2 = HIGH. The 16×2 I2C LCD correctly displayed the status CLOCKWISE, ANTI-CLOCKWISE or STOPPED.

## PRECAUTIONS

1. Check the pin labels printed on the module (IN1, IN2, ENA, OUT1, OUT2, VS, GND) before powering.
2. Always join the grounds of Arduino and the motor supply.
3. Do not connect the motor directly to Arduino pins.
4. Do not exceed the rated voltage/current of the motor and the module.
5. Connect the LCD SDA to A4 and SCL to A5 correctly; a wrong I2C address or wiring gives a blank display.
6. Switch off the supply before changing any connection.